

Technical Summary:

BoNBoN Alignment for Large Language Models and the Sweetness of Best-of-n Sampling

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1 Research Problem and Motivation

1.1 Problem Setting

Alignment of large language models refers to adapting a base language model so that its generated responses are more likely to satisfy a desired preference criterion (for instance, being more helpful, harmless, or higher-quality according to a ranking system), while avoiding unintended changes to other properties of the output. The core tension is a trade-off between:

- **Target improvement:** producing responses that score higher under a preference or reward signal.
- **Off-target stability:** avoiding large drift in other attributes of the output distribution (such as verbosity or stylistic changes).

1.2 Prior Limitations and Research Gap

Many alignment pipelines introduce **explicit regularization** that attempts to control how far the aligned model deviates from the base model. A common formal proxy is the Kullback–Leibler (KL) divergence between the aligned policy and the reference policy. However, there are two core issues:

- **Mismatch between KL and salient drift:** controlling estimated KL does not necessarily control attributes that matter to humans, such as response length.
- **High-dimensional estimation risk:** KL drift must be estimated from finite samples in a very high-dimensional space, and explicit optimization against an estimate can be exploited when the estimate is inaccurate.

A separate practical limitation concerns **best-of-n sampling** at inference time. Best-of-n often yields strong alignment behavior by sampling n candidate responses and selecting the one ranked highest. However, drawing n samples per prompt increases inference cost substantially, creating a need for a training-time method that **mimics best-of-n** without requiring multi-sample inference.

2 Related Work

2.1 Preference-Based Alignment

Preference-based alignment methods learn from pairwise comparisons between two responses for the same prompt. A typical pipeline involves:

- **Reward modeling:** training a reward model (or ranker) that assigns higher scores to preferred responses.

- **Policy optimization:** optimizing a language model policy to increase the probability of responses that are preferred under the reward model.

2.2 RLHF and Contrastive Alignment Objectives

Reinforcement learning from human feedback (RLHF) uses a reward model and policy optimization with explicit regularization to the reference policy. More recent contrastive objectives, such as Direct Preference Optimization (DPO) and Implicit Preference Optimization (IPO), optimize likelihood ratios on chosen-versus-rejected pairs. These methods typically include a temperature or scaling hyper-parameter that influences the trade-off between reward improvement and drift.

2.3 Best-of-n Sampling

Best-of-n sampling is an inference-time strategy: sample multiple candidates and keep the highest-ranked. This provides a form of **implicit** drift control because the policy is not directly optimized against an explicit KL estimate. The key question is how to retain best-of-n benefits while avoiding its inference cost.

3 Dataset Construction

3.1 Datasets and Filtering

Two tasks are studied:

- **Single-turn dialogue generation** using the Anthropic Helpful and Harmless (HH) dataset.
- **Text summarization** using the OpenAI TL;DR dataset.

Due to computational constraints, the datasets are filtered to reduce sequence length. Table 1 summarizes the resulting training data.

Table 1: Datasets, sources, sizes, and preprocessing.

Dataset	Task	Training Instances	Prompts	Preprocessing and Notes
Anthropic Helpful and Harmless (HH)	Single-turn dialogue generation	106,754	Not reported	Keep prompts with response length < 500 characters. Each instance includes two model-generated responses and a preference label.
OpenAI TL;DR	Text summarization	92,831	14,764	Discard instances where input post length < 90 characters. Each instance includes two model-generated responses and a preference label.

3.2 Ground-Truth Ranker for Evaluation

To compare methods by a consistent win-rate criterion, an off-the-shelf reward model is used as a **ground-truth ranker**. Human preference labels are relabeled using this ranker so that the training and evaluation comparisons reflect the ranker-defined preference signal.

4 Query Protocol and Task Definitions

4.1 Core Symbols and Objects

The following objects define the interaction and learning setting:

- x : a **prompt** (input text).
- y : a **response** (generated text conditioned on x).
- $\pi_0(y \mid x)$: the **reference policy**, meaning the base language model distribution over responses.
- $\pi(y \mid x)$: a generic **aligned policy** to be learned.
- $r(x, y)$: a **reward or ranking score** assigned to response y under prompt x .

4.2 Win Rate as a Primary Metric

Win rate measures how often the aligned policy produces a response that is preferred over a response sampled from the reference policy.

$$p_{\pi \succ \pi_0} = \mathbb{P}_{x \sim D, Y \sim \pi(\cdot \mid x), Y_0 \sim \pi_0(\cdot \mid x)} \left(r(x, Y) \geq r(x, Y_0) \right). \quad (1)$$

Explanation of (1). D denotes the distribution over prompts. Y is a response drawn from the aligned policy $\pi(\cdot \mid x)$, and Y_0 is a response drawn from the reference policy $\pi_0(\cdot \mid x)$. The event $r(x, Y) \geq r(x, Y_0)$ means the ranker considers Y at least as good as Y_0 . The probability of this event is the **win rate**. This metric is used to quantify alignment improvement relative to the reference model.

4.3 KL Divergence as a Drift Proxy

KL divergence measures how much the aligned distribution differs from the reference distribution.

$$D_{\text{KL}}(\pi \parallel \pi_0) = \mathbb{E}_{x \sim D} \left[\int \pi(y \mid x) \log \left(\frac{\pi(y \mid x)}{\pi_0(y \mid x)} \right) dy \right]. \quad (2)$$

Explanation of (2). $D_{\text{KL}}(\pi \parallel \pi_0)$ is the Kullback–Leibler divergence from π to π_0 . The integral ranges over all possible responses y . The ratio $\pi(y \mid x)/\pi_0(y \mid x)$ measures how much more or less likely π makes y compared to π_0 . Taking the logarithm and averaging under π yields a scalar drift measure. In this work, KL is treated as an **off-target change proxy**, but it is also argued that KL may not fully reflect salient changes such as response length.

5 Modeling Approach

5.1 A General Class of Reward-Weighted Alignment Policies

A broad family of alignment policies can be written as reweighting the reference policy by a non-decreasing function of reward:

$$\pi_r(y \mid x) \propto f_x(r(x, y)) \pi_0(y \mid x), \quad \text{where } f_x \text{ is non-decreasing in its argument.} \quad (3)$$

Explanation of (3). π_r denotes an aligned policy induced by reward weighting. The function $f_x(\cdot)$ assigns larger weights to higher-reward responses and may depend on the prompt x . The proportionality symbol indicates that the right-hand side must be normalized to sum (or integrate) to 1 over y . This formulation unifies multiple alignment strategies by expressing them as **reward-based reweightings** of the reference distribution.

5.2 Win-Rate versus Drift as an Optimization Frontier

The alignment goal can be formalized as maximizing win rate subject to a fixed KL divergence level:

$$\max_{\pi} \mathbb{E}_{x \sim D} \mathbb{P}_{Y \sim \pi(\cdot \mid x), Y_0 \sim \pi_0(\cdot \mid x)}(r(x, Y) \geq r(x, Y_0)) \quad \text{subject to } D_{\text{KL}}(\pi \parallel \pi_0) = d. \quad (4)$$

Explanation of (4). π is the decision variable (the aligned policy). d is a fixed target KL value. This constrained objective captures the central trade-off: improve preference win rate while controlling distributional drift from the reference model.

5.3 Best-of-n Sampling Policy

Best-of-n sampling defines a policy by sampling n candidates from the reference policy and selecting the one with the highest reward score. Under this construction, best-of-n becomes a particular member of the reward-weighted family in (3).

A key theoretical result gives closed-form expressions for win rate and KL divergence under best-of-n:

$$p_{\pi_r^{(n)} \succ \pi_0 \mid x} = \frac{n}{n+1}. \quad (5)$$

Explanation of (5). $\pi_r^{(n)}$ denotes the best-of-n policy induced by selecting the highest-ranked sample among n draws from $\pi_0(\cdot \mid x)$. The quantity $p_{\pi_r^{(n)} \succ \pi_0 \mid x}$ is the **context-conditional** win rate, meaning win rate conditioned on a fixed prompt x . The expression $n/(n+1)$ shows that increasing n monotonically increases win rate, with diminishing returns as n grows.

$$D_{\text{KL}}(\pi_r^{(n)} \parallel \pi_0 \mid x) = \log(n) - \frac{n-1}{n}. \quad (6)$$

Explanation of (6). $D_{\text{KL}}(\pi_r^{(n)} \parallel \pi_0 \mid x)$ is the context-conditional KL divergence, meaning KL for a fixed prompt x . The term $\log(n)$ grows with n , and the subtraction $(n-1)/n$ slightly reduces the value. This equation quantifies how best-of-n increases drift as n increases.

Table 2 instantiates (5) and (6) for common n values using natural logarithms.

Table 2: Closed-form best-of-n win rate and KL values from (5) and (6).

n	Win Rate $n/(n+1)$	KL $\log(n) - (n-1)/n$
2	0.6667	0.1931
4	0.8000	0.6363
8	0.8889	1.2044

5.4 Near-Optimality of Best-of-n on the Frontier

The best-of-n policy is shown to be essentially optimal on the win-rate versus KL frontier: the maximum win-rate difference between the optimal policy and best-of-n (at $n = 2$) is less than 1 percentage point, and larger n values approximate the optimal policy even more closely.

5.5 BoNBoN: Best-of-n Fine-Tuning to Avoid Multi-Sample Inference

Best-of-n is attractive but expensive at inference time because it requires n samples per prompt. The central idea is to **train a single policy** that mimics best-of-n so that inference remains single-sample.

Two component training objectives are introduced:

- **SFT-BoN**: a supervised objective that pushes the model toward the best-of-n samples.
- **IPO-BoN**: an implicit preference objective defined on the best-versus-worst samples, designed to improve data efficiency but susceptible to a form of likelihood-ratio cheating.

BoNBoN combines them:

$$L_{\text{BoNBoN}}(\pi_\theta; \pi_0) = \alpha L_{\text{SFT-BoN}}(\pi_\theta; \pi_0) + (1 - \alpha) L_{\text{IPO-BoN}}(\pi_\theta; \pi_0). \quad (7)$$

Explanation of (7). π_θ is the trainable policy parameterized by model parameters θ . π_0 is the reference model. $L_{\text{SFT-BoN}}$ is the supervised best-of-n objective, and $L_{\text{IPO-BoN}}$ is the implicit preference objective built from best-and-worst samples. $\alpha \in [0, 1]$ is a mixing hyper-parameter that balances the two losses. This combined objective is designed so both terms target the same best-of-n policy, while the combination improves data efficiency and reduces cheating behavior.

5.6 Rationale for Combining the Two Objectives

IPO-BoN is described as more data efficient than SFT-BoN, but it can cheat by reducing likelihood of both winning and losing responses as long as the loser decreases more, shifting mass elsewhere and empirically increasing the probability of very long responses. The SFT-BoN component counteracts this by forcing high likelihood on best-of-n samples, thereby discouraging pathological probability shifts.

6 Empirical Results

6.1 Experimental Setup and Evaluation Protocol

Two tasks and two datasets are used as described in Table 1. Evaluation focuses on:

- **Win rate against the SFT baseline**, using the reward model as the judge.
- **Off-target drift indicators**, with particular emphasis on response length changes that may not be reflected by measured KL.

The reward model is also used to relabel human preference pairs so that training and evaluation align with the ground-truth ranker used in experiments.

6.2 Baselines and Hyper-Parameters

Baselines include DPO and IPO trained on both original preference pairs and best-and-worst-of-8 data variants. The hyper-parameter search spaces for β are enumerated in Appendix experimental details, and BoNBoN is trained with a default $\alpha = 0.005$ and a fixed $\beta_g^* = 0.0275482094$, with additional ablations over α and β .

Table 3: Key experimental configuration details reported in the paper.

Category	Reported Setting
Ground-truth ranker	Off-the-shelf reward model used as judge; human preferences are relabeled to match the ranker.
BoNBoN default mixing	$\alpha = 0.005$; $\beta_g^* = 0.0275482094$.
Additional BoNBoN ablations	Fixed β_g^* with $\alpha \in \{0.05, 0.2\}$ (HH) and $\alpha \in \{0.05, 0.02\}$ (TL;DR); fixed $\alpha = 0.005$ with $\beta \in \{\beta_g^*/5, \beta_g^*, 5\beta_g^*\}$.
Optimizer and training	RMSprop with learning rate 5×10^{-7} ; use the 20k checkpoint for sampling.
Hardware and runtime	Training reported on 3 A100 GPUs with 140 GB memory; approximately 3 hours to reach 20k timesteps.

6.3 Main Findings and Interpretation

The empirical narrative supports three main claims:

- **Best-of-n provides an essentially optimal trade-off** between win rate and KL divergence, with a reported maximum win-rate gap under 1 percentage point at $n = 2$, and closer approximation at larger n .
- **Explicit KL regularization can be unreliable in practice**, because estimated KL may not correspond to salient output attributes. Response length is highlighted as an attribute that can vary substantially without being reflected in measured KL.
- **BoNBoN mitigates IPO-style cheating while improving data efficiency**, by mixing SFT-BoN (which anchors probability mass on best-of-n samples) with IPO-BoN (which uses best-and-worst samples to improve sample efficiency).

6.4 Ablation Study Interpretation

The reported ablations vary α and β around the default configuration. The role of these ablations is:

- **Varying α** tests sensitivity of the balance between anchoring (SFT-BoN) and ratio-based learning (IPO-BoN).
- **Varying β** tests robustness to the effective sharpness of preference weighting around the default β_g^* scale.

7 Summary

7.1 Key Innovations versus Prior Methods

- **Implicit handling of win-rate versus drift trade-off:** best-of- n determines the trade-off through n rather than explicit KL regularization.
- **A training-time mechanism to mimic best-of- n :** BoNBoN targets the best-of- n policy as a minimizer without requiring multi-sample inference at test time.
- **A principled combination of objectives:** mixing SFT-BoN and IPO-BoN is motivated as a way to increase data efficiency while preventing likelihood-ratio cheating that can distort off-target attributes.

7.2 Limitations and Future Directions

- **Dependence on the ground-truth ranker:** experimental comparisons rely on an off-the-shelf reward model as judge, and relabeling preferences to match it. This anchors conclusions to the reward model’s notion of quality.
- **Off-target drift is multifaceted:** response length is emphasized as a salient drift dimension not captured by estimated KL, suggesting that additional drift metrics may be required for robust alignment auditing.
- **Best-of- n trade-off is still controlled by n :** while avoiding explicit KL estimation, the choice of n still governs the implicit drift level, and choosing n remains an operational design decision.